Provisional Patent Application 15 - Liana Banyan Corporation

Filing Specification — Agent-Spawn Boundary Memory, Lossless Tablet Capture, Substrate Discipline Primitives, and Retrieval-Navigation Architecture

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Provisional Patent Application 15

Liana Banyan Corporation

Applicant: Liana Banyan Corporation **Entity Type:** Wyoming C-Corporation **EIN:** 41-2797446 **Inventor:** Jonathan Jones (sole inventor) **Filing Type:** Provisional Patent Application (USPTO Patent Center) **Priority Date Goal:** 2026-04-29 (B133 fire trigger, Founder authorization) **Conversion Deadline (Prov 14 chain continuation):** April 29, 2027 **Filing Counsel (post-conversion):** Harrity & Harrity (#1) / Lloyd & Mousilli (#2)

Title of Invention: *Cooperative-Platform AI Memory Infrastructure — Agent-Spawn Boundary Pre-Injection, Lossless Vendor-Layer Tablet Capture, Corpus-Alias Registry-Keyword-Extension, and Substrate Discipline Primitives (Stone Tablet Imperative, Call Sign Addressing, Fire Control, Cut In Half / Both-Feet, and Multi-Cylinder Dispatch Architecture)*

Filed under: #2260 Cooperative Defensive Patent Pledge (Liana Banyan Corporation, B098).

PUBLICATION GATE HARD: This specification is **internal-only** (counsel review + Founder fire required). Knight scope: build and stage. Founder + counsel scope: review and authorize USPTO submission. Mechanical Fire Control per Founder directive (B132 turn ~35).

Filing Manifest — One-Line Summaries

This provisional application bundles the following innovations into one specification, organized by reduction-to-practice cluster. Each innovation is presented with its A&A formal description (where formalized) or thresh-stub description (where formal drafting is pending). All claims-language is provisional-grade; counsel polishes during the 12-month conversion window.

Innovation numbers from #2327 onward are **pre-filing candidate assignments**. Bishop assigns final canonical sequence at filing-preparation time. Where a number is marked "candidate," counsel should verify non-collision against the A&A formal ledger before conversion.

#	Title	Short Description
2327	Stone Tablet Imperative + Vendor-Layer Tablet Capture (K541)	Platform-wide lossless-append discipline: every agent output preserved as Stone Tablet; extended to vendor-API boundary so full request/response captured BEFORE vendor's own summarization-tax fires. Composites with Pheromone (#2317) for sub-ms re-retrieval. K541 call sign: `v-vendor-layer-tablet-capture-K-VTC-B132 · d57434b`.

#	Title	Short Description
2329	Sweeper Scribe (Triage Primitive)	Agent-class triage scribe that acts on Scavenger re-verification verdicts: LEAVE / ARCHIVE PILE / ANGEL OF DEATH / REFRESH disposition classes. Counterpart green-keeping pair with #2328 Scavenger Scribe.
2330	Fire Control Directive (Structural Bylaw Primitive)	Platform-wide authorization rule: only humans may pull the trigger on publication, deployment, or canonical-state-flip actions. AI agents scope, scaffold, build, test, and recommend — humans authorize. Structural Bylaw class; cannot be changed by AI vote or normal product process.
2331	Call Sign Primitive (Tag·Commit Addressing Layer)	A durable, addressable, agent-portable identifier formed by pairing a git tag with its commit SHA (`<tag> · <commit-sha>`). Serves as the substrate's L3 routing layer: TCP/IP analog for agent handoffs, pheromone transport payload, and Stone Tablet pointer across agent boundaries.
2332	Wrasse Scribe (Agent-Spawn Boundary Pre-Injection)	Pre-injects canonical answers for known rote-cognition trigger classes BEFORE the agent's first reasoning token fires, eliminating ~41–90% of agent-spawn-boundary context tax. 66-entry registry; sub-ms lookup (mean 0.059ms); Staleness filter + token cap. Empirical receipt: 41.1% proxy lower bound (K545, Phase E gate CLEARED). Call signs: `v-wrasse-scribe-mvp-K540 · 8ca55cd` / `v-wrasse-`

#	Title	Short Description
		`wiring-hardening-K544 · 429abbd` / `v-wrasse-empirical-campaign-K545 · d3802cb`.
2333	Cut In Half / Both-Feet Principle (Substrate Configuration Doctrine)	When configuring a budget cap, retry count, timeout, or threshold-class parameter: set it high enough that the cap never binds on normal-or-elevated operating variance (Cut In Half → effectively Double It). When a parameter has been silently failing or the ceiling has been hit repeatedly, jump to 10× headroom in one step (Both-Feet / Awakenings principle). Converts iterative friction-tax into a single binary configuration decision, compounding with Fire Control's binary-trigger discipline.
2334	Multi-Cylinder Dispatch Patterns (Aston Martin Pistons / Straight Six / SUNFLOWER / BUZZYBEE)	Three named parallel-dispatch architectures for cooperative AI substrate work: (1) Straight Six — 6 subagents / Miner aggregation, default coverage pattern; (2) SUNFLOWER — 10 subagents / 2-squad / dual-lens reconciliation, for corpus work requiring two framing axes; (3) BUZZYBEE — 8 subagents / 2 Agent-class riders in-flight, for highest-stakes work requiring Agent-class judgment during execution, not post-hoc. Named via Aston Martin session-piston metaphor: each session = a piston in the awareness engine; each Augur = an Aston Martin.
2335	Registry-Keyword-Extension Method +	A reproducible method for closing alias gaps in three-layer Cathedral retrieval

#	Title	Short Description
	System (Corpus-Alias Retrieval Discipline)	architecture (Scribe registry / corpus entries / scoring engine) without engine modification: (1) audit corpus-entry observation bodies for natural-language alias terms; (2) propagate aliases to Scribe-level keyword arrays in `registry.yaml`; (3) rebuild substrate; (4) empirical refire validation. Reproducibility demonstrated K473 → K547 (same mechanism, two independent applications, 100% HOT receipt on N=33 queries). Call sign: `v-retrieval-refinement-K547 · 99a4869`.

Total innovations in this filing: 7 primary innovations, 9 reduction-to-practice anchors.

Innovation count note: Items #2328 (Scavenger Scribe, B131) and previously-unfiled B127-B131 innovations (#2308, #2311, #2313-#2326) are carried in the Prov 14 conversion window or staged for a companion Prov 15 cluster. This filing covers specifically the B132-B133 architectural primitive cluster originated by Founder during sessions B132-B133 (2026-04-28/29).

Cooperative Defensive Patent Pledge (#2260) — Filing Posture

All innovations in this provisional are filed under the LB Cooperative Defensive Patent Pledge framework. Every nonprofit, cooperative, and academic institution receives use rights to the architecture **free, forever**, on nothing more than an IRS-verified EIN (or international equivalent). The Pledge is administered by Liana Banyan Corporation governance per the Structural Bylaws. This filing protects the architecture from third-party enclosure; it does not claim exclusivity over cooperative-sector use.

The Fire Control Directive (#2330) and the Stone Tablet Imperative (#2327) compose directly with the Pledge: Fire Control ensures human oversight at every publication and canonical-state-flip trigger; the Imperative ensures the full substrate record is preserved for cooperative-sector replication. Together they are the operational and archival layers of the Pledge's self-enforcing discipline.

Path B Discipline (Proof Before Claim)

All claims in this filing are grounded in empirical receipts per the Path B discipline (B130 Founder-ratified, `project_path_b_proof_before_claim_b130.md`): build the reduction-to-practice artifact first, file the provisional with the working build as Section 2 evidence.

Per-innovation empirical anchor summary:

#	Empirical Anchor	Call Sign	Receipt Class
2327	K541 vendor-tablet JSONL confirmed on disk; 12 unit tests pass	`d57434b`	ANCHORED (infrastructure)
2329	Scavenger + Sweeper scope memo ratified B132; triage classes LEAVE/ARCHIVE/ANGEL/REFRESH defined	—	THRESH-STUB (implementation pending)
2330	Fire Control operative across all K525+ sessions; mechanical `*_ENABLED=false` pattern used K525/K530/K545	multiple	ANCHORED-AS-DISCIPLINE
2331	Call Signs in active use K523-K548; Wrasse registry carries Call Sign trigger class	multiple	ANCHORED-AS-DISCIPLINE
2332	41.1% proxy lower-bound Phase E gate cleared (K545); 11/11 unit tests; sub-ms benchmark; 66-entry registry	`d3802cb`	ANCHORED (proxy lower bound)
2333	max_entries 100→1000 empirical receipt (K535 B132); PER_CONDITION_PAUSE 100→1000 (B132 same session)	K535	ANCHORED (two independent applications)
2335	33/33 HOT = 100% (K547, $1.83 industry-term); K473 prior-art reproduction; K546 Phase A intermediate	`99a4869`	ANCHORED (100%-empirical-receipt)

Filing Window and Priority

This provisional application is filed within the priority-date window opening 2026-04-29 (B133 Founder fire trigger). The 12-month conversion deadline for this filing anchors no later than 2027-04-29. Counsel coordination is contemplated post-filing for conversion-time claim polish. The Prov 14 application (filed 2026-04-26, Priority Date 2026-04-26) is the direct predecessor; this filing continues the sequential provisional chain protecting the cooperative-platform AI memory substrate architecture.

Specification Body — Innovations in Detail

The following sections present the full description of each innovation. Each section begins with the innovation's title and short description, followed by the complete thresh-stub description (formal A&A drafting pending) or existing formal description. Method claims and system claims are both presented for patentable innovations. All claims-language is provisional-grade; counsel polishes during conversion.

Section I — Lossless Substrate Capture Architecture (B132-B133)

#2327 Stone Tablet Imperative + Vendor-Layer Tablet Capture

Innovation Title

Stone Tablet Imperative: Lossless Append-Only Agent-Output Architecture with Vendor-API-Layer Extension

Classification

- **Innovation Number:** #2327 (candidate)
- **Crown Jewel Candidate:** YES — composites Pheromone Substrate (#2317) with lossless append-only discipline across ALL agent boundary types including vendor-API boundaries, creating a substrate in which no inference output is ever discarded
- **Patent Provisional:** Prov 15
- **Date of Conception:** April 28, 2026 (B132 turn 9 — principle; turn 32 — vendor-layer extension)
- **Session of Origin:** B132
- **Conceived by:** Jonathan Jones (Founder & General Manager)
- **Reduction to Practice:** K541 (`v-vendor-layer-tablet-capture-K-VTC-B132 · d57434b`)

Description

The Stone Tablet Imperative names and formalizes the architectural discipline that every summary produced by any agent (Bishop, Knight, Pawn, subagent, Augur, Detective, Miner) must be preserved as a durable, append-only Stone Tablet — not summarized-and-discarded.

Founder framing (B132 turn 9, 2026-04-28): *"This points out the real innovation — we just need to convert all those summaries into stone tablets, across the board. AND save and store and make accessible. The underlying reason for making AI work the way it does* *is to be able to come up with empirically possible answers from enormous mounds of context — but the way it does gets enlarged and then shrunk again on summaries."*

The information-theoretic argument: AI's load-bearing value is pattern recognition over large context. Summarization is lossy compression. When an agent summary is discarded after reading, the raw signal is destroyed. When the raw output is preserved as an append-only Stone Tablet and indexed by the Pheromone Substrate (#2317), the raw signal remains available for future cross-domain transfer, future Detective queries (#2316), and future Pheromone-fast retrieval at sub-millisecond latency.

The vendor-layer extension (K541, d57434b) applies the Imperative at the vendor-API boundary — capturing the full vendor request + response BEFORE the vendor's own summarization tax fires. Vendors charge per-token on input + output. When vendors maintain rolling context windows, they bill for internal summarization. If the cooperative platform captures the raw response as a Stone Tablet at call time, the platform holds the full-fidelity record independently of the vendor's session management. On subsequent calls, the platform can inject relevant Tablets as context blocks via Pheromone-fast retrieval — the vendor processes a focused query without needing to maintain its own rolling summarization, reducing per-call token cost at the vendor layer.

Method Claim (Stone Tablet Imperative)

A method for lossless preservation of agent outputs in a cooperative-platform multi-agent AI system, comprising:

1. Receiving an output from a first agent (Bishop, Knight, Pawn, subagent, Augur, or Miner agent) operating within a Cathedral substrate architecture;
2. Writing the complete output to a durable, append-only Stone Tablet record — where "append- only" means no prior record may be modified or deleted, only new records appended;
3. Indexing the Stone Tablet via a stigmergic cross-Scribe index (Pheromone Substrate, #2317) for sub-millisecond retrieval by subsequent agents;
4. Providing the indexed Stone Tablet for retrieval by downstream agents without re-derivation of the output's content;
5. Storing the Stone Tablet at a known canonical path associated with the Call Sign of the agent session that produced it (#2331, Call Sign primitive);
6. Repeating steps 1–5 for every agent operating within the substrate, including subagents dispatched within the same session.

Claim variation — vendor-layer extension: A method as above further comprising: - Intercepting a vendor-API request-response pair at the calling agent layer; - Writing the complete request payload and complete response payload to a vendor-layer Stone Tablet prior to any summarization or truncation by the vendor's session-management system; - Indexing the vendor-layer Tablet by Pheromone Substrate topics derived from the query classification and response content; - On subsequent vendor-API calls, querying the Pheromone Substrate for relevant prior-response Tablets and injecting them as context blocks before the new request — thereby reducing per-call token count and eliminating the vendor's internal summarization overhead.

System Claim (Vendor-Layer Tablet Capture)

A system for vendor-API-layer lossless capture in a cooperative-platform AI memory infrastructure, comprising:

- A **capture module** (`vendor_tablet_capture.py`, K541 implementation) wrapping every vendor-adapter (`anthropic_adapter.py`, `openai_adapter.py`, `google_adapter.py`, `perplexity_adapter.py`, `groq_adapter.py`, `together_adapter.py`, `ollama_adapter.py`) via a context-manager pattern (`capture_vendor_call()`) that writes `{request, response, model, ts, vendor_spend_usd, mastery_context, query_class}` to an append-only JSONL tablet at `librarian-mcp/stitchpunks/data/vendor_tablets/<vendor>/<YYYY-MM-DD>.jsonl`;
- A **redaction layer** (`_redact()`) that removes API keys, tokens, and PII from vendor-layer Tablets before storage, preserving full semantic content without credential exposure;
- A **query interface** (`vendor_tablet_query()`) that retrieves prior-session Tablets by topic keys, model, or time range;
- A **Pheromone index extension** that registers vendor-layer Tablet topics in the existing `pheromone_substrate/index.jsonl`, enabling sub-millisecond cross-session retrieval;
- An **MCP tool binding** (`vendor_tablet_query` tool registered in `server.ts`) that exposes the vendor Tablet query interface to any MCP-connected AI client.

Reduction-to-Practice Evidence

- **K541 commit `d57434b`**, tag `v-vendor-layer-tablet-capture-K-VTC-B132`
- 7 adapters wired; 12 unit tests all green at commit
- JSONL confirmed landing at canonical path during test execution
- TypeScript stub `src/vendor_tablet_capture.ts` matching Python API surface
- `vendor_tablet_query` MCP tool registered and callable
- Receipt class: **ANCHORED** (infrastructure works, tests pass, JSONL on disk)
- Impact on vendor-spend reduction: **HYPOTHESIS** pending empirical campaign (predicted 40-60% additional reduction vs. vendor-native memory products)

Composes With

- #2317 Pheromone Substrate — the retrieval layer for indexed Tablets
- #2316 Detective Scribe — cross-examines Tablets for provenance and drift detection
- #2326 Sovereignty-by-Construction Reproducibility Pack — Tablets ARE the receipts
- #2331 Call Sign Primitive — every Tablet carries the Call Sign of its parent session
- #2332 Wrasse Scribe — Wrasse registry IS a Stone Tablet; Wrasse pre-injection uses Tablets as its canonical answer source

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Section II — Triage and Authorization Architecture (B132)

#2329 Sweeper Scribe (Triage Primitive)

Innovation Title

Sweeper Scribe: Substrate-State Triage Agent with Four-Class Disposition Architecture

Classification

- **Innovation Number:** #2329 (candidate)
- **Crown Jewel Candidate:** YES — closes the green-keeping loop for the Cathedral substrate: Scavenger (#2328) asks "is this still true?"; Sweeper acts on the answer with a formally defined four-class disposition, enabling substrate longevity without unbounded growth
- **Patent Provisional:** Prov 15
- **Date of Conception:** April 28, 2026 (B132 turn ~35, Founder explicit ratification: "Good")
- **Session of Origin:** B132
- **Conceived by:** Jonathan Jones (Founder & General Manager)
- **Formal Anchor:** B132 standing items disposition #3 + #5 (both ratified)

Description

Founder framing (B132, ratified turn ~35): *"Sweeper does Triage, essentially. Yes."*

The Sweeper Scribe is an agent class that operates as a **triage primitive** within the Cathedral substrate's green-keeping cycle. After the Scavenger Scribe (#2328) re-verifies a substrate entry's accuracy on its geometric-spaced longevity-index schedule, the Sweeper acts on the Scavenger's verdict by assigning a disposition class and executing the corresponding action.

The four Sweeper disposition classes:

1. **LEAVE** — the entry is still true, still load-bearing, no action needed; Scavenger longevity clock resets; no write required.
2. **ARCHIVE PILE** — the entry is historically true but no longer load-bearing for active inference; moves to an archaeology-layer JSONL outside the active Cathedral Scribe; retained for Stone Tablet Imperative compliance (never deleted, only relocated to dormant-Scribe layer).
3. **ANGEL OF DEATH** — the entry is verified false, superseded, or definitively refuted; retired from active canon; moved to a `refuted_records.jsonl` tombstone with full provenance; the active Scribe index is updated to exclude the entry from retrieval.
4. **REFRESH** — the entry's claim is still true but its framing is stale (outdated numbers, deprecated terminology, or superseded canonical reference); queued for a Bishop canonical- update session; entry remains in active retrieval with a STALE flag until the update lands.

The Scavenger + Sweeper green-keeping pair: - **Scavenger (#2328):** asks *"Is this still true?"* — refire mechanism on geometric interval - **Sweeper (#2329):** asks *"What do I do about it?"* — acts on Scavenger's answer with formal disposition

Together, Scavenger + Sweeper form the substrate's longevity-management layer: the Cathedral can grow indefinitely because entries that become false or dormant are formally retired rather than silently staling.

Method Claim (Sweeper Scribe)

A method for substrate-state triage in a cooperative-platform AI Cathedral architecture, comprising:

1. Receiving a re-verification verdict from a Scavenger Scribe (#2328) for one or more Cathedral substrate entries, where each verdict classifies the entry's current accuracy and load-bearing status;
2. For each entry, applying a four-class triage disposition:
   - a. If the Scavenger verdict is STILL-TRUE and LOAD-BEARING: assign LEAVE, reset the Scavenger longevity clock, no write required;
   - b. If the Scavenger verdict is STILL-TRUE but NO-LONGER-LOAD-BEARING: assign ARCHIVE PILE, move entry to a dormant-Scribe layer per Stone Tablet Imperative, update the active index to exclude;
   - c. If the Scavenger verdict is VERIFIED-FALSE or REFUTED: assign ANGEL OF DEATH, write a tombstone record to `refuted_records.jsonl` with full provenance (source session, date of refutation, Call Sign of refuting evidence), update active index to exclude;
   - d. If the Scavenger verdict is STALE-FRAMING: assign REFRESH, mark entry STALE in active retrieval, queue a canonical-update session;
3. Writing a Sweeper disposition record as an append-only Stone Tablet (per #2327 Stone Tablet Imperative) containing the entry ID, disposition class, Scavenger verdict, timestamp, and the Call Sign of the Sweeper dispatch session;
4. Reporting disposition summary to the Bishop substrate-management layer.

System Claim

A system for cooperative-platform AI substrate longevity management, comprising:

- A **Scavenger Scribe subsystem** (#2328) that monitors Cathedral substrate entries on geometric-spaced longevity-index schedules and produces STILL-TRUE / NO-LONGER-LOAD-BEARING / VERIFIED-FALSE / STALE-FRAMING verdicts;
- A **Sweeper Scribe subsystem** (#2329) that receives Scavenger verdicts and applies the four-class disposition protocol (LEAVE / ARCHIVE PILE / ANGEL OF DEATH / REFRESH);
- A **dormant-Scribe repository** (Catacombs, #2285) that receives ARCHIVE-PILE and ANGEL-OF-DEATH entries per the Stone Tablet Imperative (no deletion, only relocation with provenance);
- A **STALE-flag retrieval layer** that surfaces entries marked REFRESH in active retrieval with a canonical staleness indicator until a Bishop update lands;
- A **disposition Stone Tablet** (`sweeper_dispositions.jsonl`) appended with every triage action for retroactive audit and cross-session observability.

Composes With

- #2285 Catacombs — dormant-Scribe repository receives ARCHIVE PILE entries
- #2316 Detective Scribe — Sweeper can trigger Detective investigation before ANGEL OF DEATH

- #2326 Reproducibility Pack (#2326) — disposition records ARE reproducibility receipts
- #2327 Stone Tablet Imperative — Sweeper dispositions are Stone Tablets; tombstones preserved per the Imperative
- #2330 Fire Control — ANGEL OF DEATH action is a canonical-state-flip requiring human review before execution on Crown Jewel or production-canonical entries

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#2330 Fire Control Directive (Structural Bylaw Primitive)

Innovation Title

Fire Control: Human-Authorization-Required Structural Bylaw for AI Cooperative Platforms

Classification

- **Innovation Number:** #2330 (candidate)
- **Crown Jewel Candidate:** YES — structural governance primitive with no known prior art for a Structural Bylaw class rule in a cooperative AI platform that formally restricts AI agents from autonomous execution of publication, deployment, or canonical-state-flip actions
- **Patent Provisional:** Prov 15
- **Date of Conception:** April 28, 2026 (B132 turn ~35, Founder ratification: "Good")
- **Session of Origin:** B132
- **Conceived by:** Jonathan Jones (Founder & General Manager)
- **Reduction to Practice:** Operational across K525/K530/K532/K545 (mechanical `*_ENABLED=false` pattern); `knowledge_pump.json` `KNOWLEDGE_PUMP_TEST_ENABLED` flag; all Wave 1 + Crown Letter dispatch gates

Description

Founder framing (B132 turn ~35): *"Hard Founder directive, platform-wide: Only humans can authorize 'actually doing it now going live' actions. AI agents scope, scaffold, build, test, verify, audit, and recommend — but the trigger to fire publication, dispatch, deployment, or canonical-state-flip is reserved to the Founder hand alone (or designated human authority)."*

The Fire Control Directive is a **Structural Bylaw class rule** in the Liana Banyan cooperative-platform AI governance architecture. It formally partitions the responsibilities of AI agents and human principals in a multi-agent cooperative AI system:

AI agent scope (unrestricted): scope, scaffold, design, build, test, verify, audit, recommend, draft, simulate, stage, and report.

Human authorization required (restricted actions): - Publishing or sending any content to a public-facing surface (website, email, social, USPTO submission, regulatory filing) - Deploying any build to a production environment accessible to members - Flipping a canonical-state record (canonical innovation count, patent count, member count) from one value to another - Submitting any provisional or utility patent application to USPTO or equivalent authority - Dispatching any message to a Crown Letter addressee or Wave 1/2/3 launch recipient

Naming origin (Founder B132): “Fire Control” — military and aviation register matching Founder’s Infantry 11B + Aviation 15A FAA Commercial Rotary Wing background. Fire Control systems in aviation and weapons management use exactly this authorization structure: the firing circuit is physically completed only when an authorized human action provides the final arming and release. An automated system may track, aim, calculate, and recommend — but the trigger is a human hand.

Composition with Cut In Half / Both-Feet (#2333): Fire Control says “humans pull the trigger.” Cut In Half / Both-Feet says “configure the substrate so the human-fire decision is binary (fire / don’t fire), not iterative (bump-bump-bump-fire).” Together: thresholds are set with enough headroom that the human authorization moment is a single binary decision, not rubber-stamping of incremental bumps.

Mechanical enforcement pattern (K525 origination, applied uniformly since): Every deployable artifact is built with a feature-flag gate defaulted `false`:

```
KNOWLEDGE_PUMP_TEST_ENABLED=false (knowledge_pump.json)
OMNIBOX_EXTENSION_PUBLISHED=false (background.js, K530)
CONDUCTOR_BATON_ENABLED=false (K525)
WAVE_1_DISPATCH_AUTHORIZED=false (future dispatch gate)
USPTO_SUBMISSION_AUTHORIZED=false (this filing gate)
```

The AI agent builds the full artifact in production-ready form. The flag remains `false` until the Founder (or designated human authority) explicitly authorizes and flips it.

Method Claim

A method for human-authorized execution control in a cooperative-platform multi-agent AI system, comprising:

1. Maintaining an AI agent task taxonomy partitioned into (a) UNRESTRICTED tasks (scope, build, test, verify, audit, recommend, stage) and (b) FIRE-GATED tasks (publish, deploy, file, dispatch, flip canonical state);
2. For every FIRE-GATED task output, generating a production-ready artifact WITH a mechanical authorization gate defaulted to a disabled state (`FEATURE_ENABLED=false` or equivalent);
3. Requiring a positive human authorization action to transition the gate from disabled to enabled before the output reaches any external-facing surface;
4. Preserving the authorization event as a Stone Tablet record (per #2327) containing the principal identity, timestamp, Call Sign of the artifact being authorized, and the authorization scope;
5. Ensuring the authorization gate is non-bypassable by AI agent action — the mechanical gate (source-level boolean, config file entry, or equivalent) requires a human-initiated write to flip.

Composes With

- #2327 Stone Tablet Imperative — every Fire Control authorization event is a Stone Tablet
- #2329 Sweeper Scribe — ANGEL OF DEATH triage on Crown Jewel entries requires human review
- #2331 Call Sign Primitive — authorization events cite the Call Sign of the authorized artifact
- #2332 Wrasse Scribe — Wrasse registry does not pre-inject authorization bypass paths
- #2333 Cut In Half / Both-Feet — together form the binary-decision-at-single-threshold compound
- #2260 Cooperative Defensive Patent Pledge — Fire Control IS the enforcement layer for the Pledge's "only Founder or governance fires" requirement

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Section III — Addressing and Agent-Boundary Primitives (B132)

#2331 Call Sign Primitive (Tag·Commit Addressing Layer)

Innovation Title

Call Sign: Durable Cross-Agent Substrate Addressing Via Tag-Commit SHA Pairs

Classification

- **Innovation Number:** #2331 (candidate)
- **Crown Jewel Candidate:** YES — establishes the substrate's L3 routing layer for cross-agent communication; prior art (git tags, commit SHAs) exists but their formal composition as a named "addressing primitive" for cooperative AI agent handoffs has no known prior art
- **Patent Provisional:** Prov 15
- **Date of Conception:** April 28, 2026 (B132 turn 14, Founder-named)
- **Session of Origin:** B132
- **Conceived by:** Jonathan Jones (Founder & General Manager)
- **Formal Anchor:** `project_call_sign_tag_commit_b132.md` (Bishop memory, B132)

Description

Founder framing (B132 turn 14, 2026-04-28): *"Tag · commit is a new Call Sign. … Means like Tag - You're It. and can be the Nervous System for transferring data.. like TCP/IP … which uses pheromones… in concert."*

A **Call Sign** is the durable, addressable, cross-agent identifier formed by pairing a git tag with the commit SHA that anchors it:

<tag> · <commit-sha>

Examples (B132-B133 active record): - v-wrasse-scribe-mvp-K540 · 8ca55cd - v-vendor-layer-tablet-capture-K-VTC-B132 · d57434b - v-retrieval-refinement-K547 · 99a4869 - v-harder-panel-K548 · 34c921c

The Call Sign serves three composed functions named by Founder in B132:

1. Tag-You're-It (token passing): When Knight completes a session and lands a Call Sign, Knight has tagged the next agent. Bishop reads the Call Sign and picks up the handoff. When Bishop closes and lands its closeout Call Sign, Founder is tagged (Fire Control authorization). The Call Sign IS the token in tag-you're-it token-passing handoffs across the agent lineage.

2. Nervous System / TCP/IP (data transport addressing): The substrate's addressing layer. Call Sign is the address: <tag> = what (scoped human-readable name) + <commit-SHA> = when (cryptographically unique substrate-state snapshot). Any agent can route a query to a specific Call Sign and receive deterministic substrate-state matching that exact moment. Equivalent to L3 routing in TCP/IP: where (namespace), what (service), when (sequence).

3. Pheromones in concert (substrate transport): Call Signs are carried through the Pheromone Substrate (#2317) as topic keys. When a query lands at the Pheromone index, Call Sign references appear as addressable targets. Cross-Cathedral routing via inbound_pheromones.jsonl carries Call Sign references across Bishop/Knight/Pawn boundaries. Pheromones in concert = the transport medium; Call Signs = the packets being transported.

Method Claim

A method for durable cross-agent addressing in a cooperative-platform multi-agent AI system, comprising:

1. At completion of each agent session producing a versioned artifact, generating a Call Sign by pairing a human-readable tag name (following a canonical format, e.g., v-<description>-K<integer>) with the commit SHA of the version-controlled repository state at the time of session completion;
2. Recording the Call Sign in the agent session's handoff record (Stone Tablet, per #2327) so that subsequent agents can address the session's output by Call Sign reference;
3. Indexing the Call Sign in the Pheromone Substrate (#2317) as a topic key, enabling sub-millisecond retrieval of the session's artifacts by subsequent agents;
4. Transmitting Call Sign references (not full artifact content) across agent boundaries via the Pheromone transport mechanism — where each receiving agent can retrieve the artifact by Call Sign reference on demand;
5. Verifying a Call Sign's integrity by checking that the tagged commit SHA matches the repository's cryptographic record — thereby confirming that the addressed substrate state has not been modified since the Call Sign was issued.

System Claim

A system for cooperative-platform AI agent addressing, comprising:

- A **Call Sign registry layer** within the Wrasse Scribe (#2332) `wrasse_registry.jsonl` that maps known Call Signs to their canonical artifact locations and session contexts, enabling pre-injection of Call Sign resolutions at agent-spawn boundaries;
- A **Pheromone Substrate topic index** (#2317) that includes Call Sign references as first-class topic keys alongside domain-content topics;
- A **handoff file pattern** (Stone Tablet, #2327) where every agent session closes with a handoff record citing the session's Call Sign and the Call Signs of any sessions it depends upon;
- A **verification discipline** where agent-spawn boundary processing resolves received Call Signs against the repository's cryptographic record before acting on the referenced artifact's content.

Active Call Sign Lineage (K523-K548)

Call Sign	Session	Description
`v-pheromone-substrate-K523 · fe6bde1`	K523	Pheromone Substrate durable build (#2317)
`v-r11v2-rich-fact-indexing-K535 · 5b26d7a`	K535	Rich-fact indexing + max_entries fix
`v-wrasse-scribe-mvp-K540 · 8ca55cd`	K540	Wrasse Scribe MVP (66 entries, sub-ms)
`v-vendor-layer-tablet-capture-K-VTC-B132 · d57434b`	K541	Stone Tablet vendor-layer extension
`v-wrasse-wiring-hardening-K544 · 429abbd`	K544	Wrasse staleness filter + token cap
`v-wrasse-empirical-campaign-K545 · d3802cb`	K545	41.1% Phase E gate cleared
`v-retrieval-refinement-K547 · 99a4869`	K547	100% HOT receipt (N=33)
`v-harder-panel-K548 · 34c921c`	K548	KP Test 4 +25pp Reading-C lift

Composes With

- #2317 Pheromone Substrate — Call Signs are Pheromone topic keys
- #2327 Stone Tablet Imperative — every Tablet carries its parent Call Sign
- #2332 Wrasse Scribe — Wrasse registry includes Call Sign trigger class

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Section IV — Agent-Spawn Boundary Pre-Injection (B132-B133)

#2332 Wrasse Scribe (Agent-Spawn Boundary Pre-Injection Architecture)

Innovation Title

Wrasse Scribe: Pre-Injection of Canonical Resolutions at Agent-Spawn Boundaries for Rote-Cognition Tax Elimination

Classification

- **Innovation Number:** #2332 (candidate)
- **Crown Jewel Candidate:** YES — first known architecture that targets the agent-SPAWN BOUNDARY layer (not within-session reasoning) for systematic rote-cognition tax elimination via pre-injection of canonical resolutions before the agent's first reasoning token fires
- **Patent Provisional:** Prov 15 (Founder explicit directive B132 turn 33: "I want to get to that as fast as possible. Because it goes in Provisional 15.")
- **Date of Conception:** April 28, 2026 (B132 turn 33)
- **Session of Origin:** B132; MVP implementation K540; wiring K544; empirical K545
- **Conceived by:** Jonathan Jones (Founder & General Manager)
- **Reduction to Practice:** K540 (`8ca55cd`) + K544 (`429abbd`) + K545 (`d3802cb`)

Description

Founder framing (B132 turn 33, 2026-04-29): *"It would be even BETTER, if we got that info that Knight is processing, BEFORE Knight got it, so we can input facts. Like, all the things our toolsmith scribe picks up that we already know EVERY instance of reasoning is going to say 'Hmm. K526 isn't real words. Must be a clue. Oh it's a session iterative prompt. Oh wait, maybe I can search all the folders to see if there is a K526 somewhere…' and instead, BEFORE even 'Hmmm' happens, our new scribe that has to do this well… Wrasse Scribe, darts in and cleans that up nice to say 'here is the full path prompt instructions'. So that whole thing never happens. That will save 90% right there since it happens EVERY TIME. I want to get to that as fast as possible. Because it goes in Provisional 15."*

The naming metaphor: Wrasse fish are cleaner-class reef inhabitants that dart in to remove parasites from other fish — even predators — before infection sets in. Wrasse's value is preventive: remove the load before it compounds. The Wrasse Scribe darts in BEFORE the rote-cognition tax fires and removes the parasite (the repeated derivation cost) before the agent burns context budget on it.

The empirical problem: Every agent-spawn boundary incurs a rote-cognition tax — the agent re-derives the same path-resolution, session-ID interpretation, vocabulary definition, and canonical-number lookup that every prior session already derived. These re-derivation steps are structurally identical across sessions; they consume context and produce no novel output.

Measured re-derivation overhead (B132 empirical anchor): K-session opening typically requires 4 sequential steps before any substantive work begins: 1. “K526 isn’t real words. Must be a clue.” 2. “Oh it’s a session iterative prompt.” 3. “Maybe I can search all the folders to see if there is a K526 somewhere…” 4. “Found it at PROMPT_KNIGHT_K526_*.md.”

Steps 1-4 consume 22–90% of the opening context budget. The Wrasse Scribe pre-injects the resolution — “K526 = Knight Session 526; prompt at [full path]; here is the content” — before step 1 fires, eliminating the entire re-derivation chain.

Trigger classes (66-entry registry, K540): - `file_path` (10 entries): canonical file paths for KNIGHT_QUEUE.md, AGENTS.md, SDS.env, etc. - `vocabulary` (15 entries): Founder-coined terms (BRIDLE, SUNFLOWER, BUZZYBEE, FlutterBy, Aston Martin, Cathedral Effect, etc.) - `k_prefix` (10 entries): recent K-session identifiers → full path resolutions - `ts_prefix` (10 entries): Toolsmith friction entries → canonical recipe resolutions - `call_sign` (6 entries): recent Call Signs → commit content and landing context - `canonical_number` (5 entries): innovation count, patent counts, platform stats

Empirical receipt (K545, Phase E gate):

Metric	Baseline	Wrasse-on	Delta
`wrasse_injectable_tokens`	718	1,491	+773
`wrasse_matches`	7	15	+8
`wrasse_reduction_pct_estimated`	41.1%	55.0%	+13.9pp
K539 model estimate	—	—	90.9% (hypothesis)
Phase E gate (≥40% lower bound)	—	**CLEARED**	—

Receipt class: **ANCHORED** (proxy lower bound); 90.9% Founder claim = HYPOTHESIS (Phase F substrate-instrumentation measurement needed for publication-grade).

Phase E Architecture Decisions (K540/K544 ratified): - D.1 Trigger detection: **Regex-first MVP RATIFIED.** Sunset: embedding fallback enters Phase 2 only after measured miss-rate > 15% across K541-K545. - D.2 Pre-injection delivery: **Bishop hook / Knight + Pawn prepend-to-prompt RATIFIED.** - D.3 Activation loop: **SessionEnd hook (Option α) RATIFIED.** `bishop_session_end.py` deployed; extracts K-NNN / TS-NNN / Call Sign tokens; writes state file; SessionStart reads and injects. Activation loop closed K544.

Phase E Conditions (status as of K544/K545): 1. Staleness expiry rule (30d / <3 verify) — **MET (K544)** 2. Injection size cap (MAX_INJECTION_TOKENS=2000) — **MET (K544)** 3. Publication gate — **FOUNDER FIRE PENDING** (gates on Phase F ≥40% measured)

Method Claim

A method for rote-cognition tax elimination at agent-spawn boundaries in a cooperative-platform multi-agent AI system, comprising:

1. Maintaining a registry of known rote-cognition trigger classes — patterns that an agent is known to re-derive identically across sessions — organized by trigger type (session-ID prefix, vocabulary term, file path class, Call Sign pattern, canonical number);
2. At each agent-spawn boundary (Knight K-session start, Pawn dispatch start, Bishop session open, subagent dispatch), scanning the incoming agent context for trigger-class patterns using a regex-first lookup against the registry;
3. For each matched trigger, retrieving the canonical pre-resolved answer from the registry (e.g., "K526 → Knight Session 526; prompt at [full path; here is full content]");
4. Injecting the pre-resolved canonical answers into the agent's context BEFORE the agent's first reasoning token fires, replacing the rote-derivation steps with the canonical resolution;
5. Enforcing an injection size cap (MAX_INJECTION_TOKENS) to prevent injection-block from becoming a context burden larger than the rote-cognition tax it eliminates;
6. Applying a staleness filter to remove registry entries older than a freshness threshold with insufficient re-verification count, ensuring only still-valid resolutions are injected;
7. At each agent-session boundary, detecting new K-NNN / TS-NNN / Call Sign tokens in the session's output and appending new registry entries for future pre-injection.

Claim variation — empirical-gate condition: The method of claims 1-7, wherein publication and external deployment of the registry and injection mechanism are gated behind an empirical Phase E measurement confirming ≥40% rote-cognition tax reduction in at least one session class under the Stone Tablet Imperative.

System Claim

A system for agent-spawn boundary pre-injection in a cooperative-platform AI infrastructure, comprising:

- A **Wrasse registry** (`wrasse_registry.jsonl`) — append-only JSONL containing entries with fields `{trigger_class, trigger_pattern, canonical_resolution, last_verified_ts, verification_count, source_session}`;
- A **lookup module** (`wrasse_lookup.py`) with regex-first pattern matching achieving sub-millisecond mean lookup latency (demonstrated mean 0.059ms / p95 0.066ms / 100% < 1ms across 1,000 benchmark runs);
- An **injection module** (`wrasse_inject.py`) that composes matched resolutions into a pre-injection block, enforces MAX_INJECTION_TOKENS=2000, and applies oldest-verified-first eviction when cap is reached;
- A **hook module** (`wrasse_hook_ext.py`) providing API to Bishop session-start and Knight K-prompt-prelude callsites;
- A **measurement module** (`wrasse_measure.py`) that instruments session context to compute `wrasse_injectable_tokens`, `wrasse_matches`, and `wrasse_reduction_pct_estimated`;
- A **SessionEnd hook** (`bishop_session_end.py`) that extracts trigger tokens from completed session transcripts and appends new registry entries;
- A **Stone Tablet session ledger** (`session_ledger.jsonl`) preserving per-session measurement records per the Stone Tablet Imperative (#2327), including invalidated records.

Composes With

- #2316 Detective Scribe — handles UNKNOWN trigger classes; feeds Wrasse registry
- #2317 Pheromone Substrate — Wrasse uses Pheromone for fast registry topic lookup
- #2327 Stone Tablet Imperative — Wrasse registry IS a Stone Tablet; every injection is itself a Stone Tablet entry
- #2331 Call Sign Primitive — every Call Sign is a Wrasse-resolvable trigger
- #2326 Reproducibility Pack — Wrasse session_ledger carries reproducibility anchors

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Section V — Substrate Configuration Doctrine (B132)

#2333 Cut In Half / Both-Feet Principle (Substrate Configuration Doctrine)

Innovation Title

Cut In Half / Both-Feet: Binary-Headroom Threshold Configuration Doctrine for Cooperative-Platform AI Infrastructure

Classification

- **Innovation Number:** #2333 (candidate)
- **Crown Jewel Candidate:** Structural-principle class; composites with Fire Control (#2330) to form the complete "configure once, fire once" binary-decision substrate discipline
- **Patent Provisional:** Prov 15
- **Date of Conception:** April 28, 2026 (B132 turn 8 — Cut In Half; turn 11 — Both-Feet Awakenings principle; both Founder-named and empirically demonstrated within same B-session)
- **Session of Origin:** B132
- **Conceived by:** Jonathan Jones (Founder & General Manager)
- **Reduction to Practice:** K535 `max_entries` 100→1000 (same B-session); `PER_CONDITION_PAUSE_USD` raised to B132 headroom; Pawn per-dispatch caps ($1→$5 / $10→$50)

Description

Founder framing (B132 turn 8): *"Let's go ahead and make it 1. So I don't EVERY have to worry about this capped mess. … I prefer the Cut In Half principle, which, inverted, as usual, means Double it."*

The Cut In Half principle: When configuring a budget cap, retry count, timeout, or threshold-class parameter, set it high enough that the cap never binds on normal-or-elevated operating variance. The bump-cap-and-resume cycle is itself friction — usually more expensive in Founder time and flow-state cost than the dollars the cap protects against. Configure to ~10× worst-observed, not 1.2× estimated.

The Both-Feet corollary (Awakenings principle — B132 turn 11): Founder framing: *"This reminds me of that movie with Robin Williams and Robert De Niro… Awakenings, where Robin Williams is the Doctor, and he acts on his theory, and it only works when he uses an insane-at-the-times-in-conventional-wisdom amount of a drug that… had truly astounding results. Sometimes, you need to jump with BOTH FEET."*

When a parameter has been silently failing OR the cap has been hit more than once, do not titrate — jump to 10× headroom in one configuration change. The third bump is the proof you titrated when you should have jumped.

The empirical receipt (same B-session as principle naming): K535 closeout reported the load-bearing bug as `max_entries: 100 → 200` in two Cathedral adapters. The Both-Feet principle was named in B132 turn 11; immediately applied to the SAME parameter as turn 13: `max_entries 200 → 1000`. The cap had silently truncated RC (entries 100-124) and HP (entries 125-149) for every benchmark run since K471. The Both-Feet move on max_entries eliminated the structural truncation rather than re-incurring it at 200. **The principle's empirical proof landed inside the same B-session in which the principle was named.**

Application classes: - **Budget caps:** set to ~10× worst-observed - **Retry counts:** set to self-resolve transient failures - **Timeouts:** set to ~5-10× expected wallclock - **Corpus capacity caps:** `max_entries` — Both-Feet to 10× current corpus size - **Spend gates:** `PER_CONDITION_PAUSE_USD` — cut the friction; both feet at $1000

Method Claim

A method for threshold configuration discipline in a cooperative-platform AI infrastructure, comprising:

1. For any configurable threshold parameter (budget cap, retry count, timeout, corpus size limit) subject to operational binding — identifying the worst-observed operational value for that parameter across N prior sessions;
2. Setting the threshold to at least 10× the worst-observed value (Cut In Half / Double It), ensuring the cap does not bind on normal-or-elevated operating variance;
3. When a cap has been incremented more than once in response to operational binding (the titration pattern), triggering a Both-Feet reconfiguration: setting the threshold to at least 10× the current operational value in a single configuration change;
4. At each threshold reconfiguration, writing a Stone Tablet record (per #2327) containing the old value, new value, empirical worst-observed, and the Call Sign of the session that triggered the reconfiguration;
5. Composing with Fire Control (#2330) so that the threshold reconfiguration is a single binary decision (fire reconfiguration / don't fire) rather than an iterative bump-authorize-bump cycle requiring repeated human authorization.

Composes With

- #2327 Stone Tablet Imperative — threshold changes are Stone Tablets

- #2330 Fire Control — binary-decision configuration per Fire Control discipline
- #2332 Wrasse Scribe — `MAX_INJECTION_TOKENS=2000` and staleness thresholds set per Both-Feet (sufficient headroom for foreseeable registry growth)

Section VI — Multi-Cylinder Dispatch Architecture (B132)

#2334 Multi-Cylinder Dispatch Patterns (Aston Martin Session Pistons / Straight Six / SUNFLOWER / BUZZYBEE)

Innovation Title

Multi-Cylinder Cooperative AI Dispatch Architecture: Named Parallel-Execution Patterns with Miner-Aggregation and Agent-Class In-Flight Judgment

Classification

- **Innovation Number:** #2334 (candidate)
- **Crown Jewel Candidate:** YES — named, parameterized, composition-aware parallel-dispatch patterns with formal selection heuristics for cooperative AI substrate work; no known prior art for the specific three-pattern architecture with formal selection heuristic and in-flight Agent-class rider pattern (BUZZYBEE)
- **Patent Provisional:** Prov 15
- **Date of Conception:** April 28, 2026 (B132: Aston Martin + Straight Six = B132 prior turns; SUNFLOWER + BUZZYBEE = B132 turn 36, Founder-named)
- **Session of Origin:** B132
- **Conceived by:** Jonathan Jones (Founder & General Manager)

Description

Founder framing (B132): *“What if we could treat each session, which REALLY, is just sparks of processing time within a set parameter of resource allowance, as pistons in the great engine of our awareness system. Then each Augur would actually have a cool car — Aston Martin, is what we could call it, because it’s a fine piece of engineering.”*

The Aston Martin metaphor (load-bearing architectural frame): - **Engine** = the cooperative AI awareness system (Cathedral + Scribes + Pheromone + Conductor) - **Piston** = a single agent session — an atomic unit of processing within fixed resource allowance (token budget + compute capacity + clock time) - **Combustion** = the work executed in a session (drafts, decisions, A&As, code) - **Crankshaft** = the Librarian — each piston’s combustion translates into rotational momentum carried to the next piston via the substrate - **Aston Martin** = each Augur-class component — the fine-engineering precision layer

The three named dispatch patterns:

Pattern 1: Straight Six (Default Coverage)

- **Layout:** 6 parallel subagents → independent Stone Tablet outputs → Miner aggregation → Bishop synthesis
- **Use case:** standard single-axis corpus coverage at moderate cost; default for most substrate work
- **Selection:** choose Straight Six when the work decomposes into parallel independent investigations of a single framing lens
- **Throughput:** 6 subagent-cylinders firing simultaneously; one Miner aggregation pass; Bishop synthesis as final step

Pattern 2: SUNFLOWER (Dual-Axis Coverage)

- **Layout:** 10 subagent juniors organized as 2 squads of 5; one Miner per squad + one reconciliation Miner; Bishop synthesis on the reconciliation output
- **Use case:** when the work requires two distinct framing lenses that each deserve dedicated coverage before reconciliation (e.g., "5 sweeps of memory cluster A + 5 sweeps of memory cluster B → cross-Miner reconciles")
- **Selection:** choose SUNFLOWER over Straight Six when two investigative axes both need full coverage, OR when corpus is too large for 6 subagents to achieve non-thin per-subagent coverage
- **Naming:** sunflower-disc geometry — many petals around a central seed-head (the Miner aggregator); two squads = two halves of the disc

Pattern 3: BUZZYBEE (High-Displacement With In-Flight Agent Judgment)

- **Layout:** 8 subagent-cylinders (Hemi-class displacement) + 2 Agent-class juniors riding alongside during execution
- **Use case:** when the work requires heavy-displacement parallel execution PLUS Agent-class judgment in-flight for synthesis, novelty detection, or anomaly handling during execution (not post-hoc via Miner)
- **Selection:** choose BUZZYBEE when Agent-class judgment must be in-flight, not post-hoc; the 2 Agent-class riders surface real-time branching opportunities and anomalies that a post-execution Miner pass would miss
- **Naming:** bumblebee aerodynamics — "shouldn't fly but does"; 8 cylinders pulling heavy weight while 2 Agents do the connective-tissue wing-work that gives the system lift
- **Cost:** highest of three patterns; reserve for highest-stakes work (Prov filing prep, Wave 1 letter cohort scrub, Cathedral re-run analysis)

Pattern selection heuristic: - Straight Six = default (good single-axis coverage, moderate cost) - SUNFLOWER = two framing lenses both need coverage before reconciliation - BUZZYBEE = Agent-class in-flight judgment required during execution

Method Claim

A method for parallel-dispatch work organization in a cooperative-platform multi-agent AI system, comprising:

1. Identifying a substrate task requiring parallel coverage by multiple agents;
2. Classifying the task against a dispatch-pattern selection heuristic:
   - a. If single framing lens: select Straight Six (6 parallel subagents);
   - b. If dual framing lens with both axes needing full coverage: select SUNFLOWER (10 subagents / 2 squads / dual-Miner);
   - c. If Agent-class in-flight judgment required: select BUZZYBEE (8 subagents + 2 Agent-class in-flight riders);
3. Dispatching the selected pattern; each subagent writes its Stone Tablet output at a canonical path under its Call Sign (per #2327 + #2331);
4. Aggregating via one or more Miner passes; the Miner reads Stone Tablets directly (no summarize-and-discard per #2327 Stone Tablet Imperative);
5. Reporting aggregated output to the Bishop synthesis layer with Call Sign references to all contributing subagent Tablets.

Composes With

- #2316 Detective Scribe — BUZZYBEE Agent-class riders may invoke Detective investigation
- #2317 Pheromone Substrate — subagent outputs indexed as Pheromone topics for cross-session retrieval
- #2327 Stone Tablet Imperative — every subagent output is a Stone Tablet; Miner reads Tablets directly
- #2330 Fire Control — Miner-aggregated outputs are staged; Founder fires publication/dispatch
- #2331 Call Sign Primitive — each subagent's output is addressed by its Call Sign

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Section VII — Retrieval Navigation Architecture (B133, K473→K547)

#2335 Registry-Keyword-Extension Method + System (Corpus-Alias Retrieval Discipline)

Innovation Title

Registry-Keyword-Extension: A Reproducible Corpus-Alias Retrieval Discipline for Three-Layer Cathedral Retrieval Architectures

Classification

- **Innovation Number:** #2335 (candidate)
- **Crown Jewel Candidate:** YES — first reproducible method with empirically-anchored 100%-HOT receipt for closing alias gaps in a three-layer retrieval architecture (Scribe registry / corpus entries / scoring engine) without engine

modification; demonstrated reproducible across two independent applications (K473, K547) with full Stone Tablet record

- **Patent Provisional:** Prov 15 (Founder explicit B133 turn 14: "absolutely does it get included in Prov 15")
- **Date of Conception:** November 2025 (K473, first application); confirmed as patentable pattern April 29, 2026 (B133 turn 14 Founder ratification of K547 reproduction)
- **Reduction to Practice:** K473 (first application, ~4 keywords, 21 of 22 failing questions cleared); K547 (`99a4869`, full audit — 14 keywords, 8 corpus entries, 33/33 HOT)
- **Call Sign:** `v-retrieval-refinement-K547 · 99a4869`

Description

Founder ratification (B133 turn 14): *"absolutely does it get included in Prov 15"*

The **registry-keyword-extension discipline** addresses the alias gap problem in multi-layer Cathedral retrieval architectures: when a user's natural-language query uses terms that semantically describe a corpus entry but do not appear in that entry's title or primary keyword array, the Scribe-level scoring engine assigns zero overlap to the correct Scribe and the query is misrouted to a lower-relevance Scribe.

The three-layer architecture where the problem occurs: 1. **Scribe registry layer** (`registry.yaml`): per-Scribe keyword arrays that the scoring engine uses for overlap calculation 2. **Corpus layer** (per-Scribe JSONL): entries with observation bodies containing natural language descriptions of the Scribe's domain knowledge — often including alias phrasings that natural-language queries would use but that don't appear in the registry keyword arrays 3. **Scoring engine layer**: computes keyword overlap between incoming query and registry arrays; routes to highest-scoring Scribe

The structural gap: observation bodies are written in natural prose and evolve over time; registry keyword arrays are curated once and become stale relative to the observation body vocabulary. The divergence accumulates silently — queries that match observation-body alias terms fail to match registry keywords and route to the wrong Scribe.

The registry-keyword-extension method: 1. Audit corpus-entry observation bodies for natural-language alias terms (phrasings that queries would use but don't appear in registry keywords) 2. Propagate identified aliases to Scribe-level keyword arrays in `registry.yaml` 3. Rebuild the substrate via `npm run rebuild` to refresh keyword matchers 4. Empirically refire the affected query subset; verify HOT rate improvement 5. Preserve pre-fix and post-fix observations as Stone Tablets per #2327

Reproducibility demonstrated: - **K473 (B118, first application):** 4 alias keywords added for AM / MJ categories; 21 of 22 previously-failing questions cleared; HOT rate: ~75% → ~96% - **K547 (B133, reproduction):** 14 alias keywords added across 8 MJ corpus entries (including MJ-07 primary fix + MJ-01/03/04/05/08 full audit); 33/33 HOT = 100%; Founder's "100% is my goal" **ACHIEVED** - **Progression chain (Stone Tablet anchored):** - Pre-K539: 75.8% HOT - Post-K539 (rich-fact expansion): 93.94% HOT - Post-K542 (grader fix): 96.97% HOT - Post-K547 (registry-keyword-extension): **100.0% HOT (33/33)**

The K473 → K547 reproduction gap (two independent applications, same mechanism, different corpus sections, separated by multiple months and personnel sessions) demonstrates that the method is a **reproducible maintenance discipline** rather than a one-off fix. This makes it patentable as a method + system rather than a single-instance engineering decision.

Composition with Wrasse Scribe (#2332): K547's keyword additions can auto-feed into the Wrasse registry once the Detective-Wrasse live-update pathway lands. Registry-keyword additions become Wrasse `vocabulary` trigger entries, providing two-layer pre-resolution: Scribe-level scoring routes to the correct Scribe (layer 1); Wrasse pre-injection pre-resolves the canonical term before the agent fires (layer 2).

Composition with Cathedral Effect (#2278): Cathedral Effect provides substrate-level uniformity (3.5pp HOT spread across 5 vendors / 23× cost spread per K535); registry-keyword- extension provides the *navigation discipline* within that substrate. Together = model- independent retrieval performance with alias-aware coverage.

Method Claim

A method for alias-gap closure in a three-layer cooperative-platform retrieval architecture, comprising:

1. Identifying query-to-Scribe misrouting events where a query's natural-language vocabulary matches the observation body of the correct Scribe but does not overlap with that Scribe's registry keyword array;
2. For each misrouted Scribe, auditing its corpus-entry observation bodies for natural-language alias phrasings — terms that accurately describe the Scribe's domain knowledge and that natural-language queries in the failed-query class would use;
3. Propagating the identified alias phrasings to the Scribe's keyword array in the registry configuration file (`registry.yaml` or equivalent), without modifying the scoring engine or corpus entry bodies;
4. Rebuilding the substrate's keyword-matcher layer to incorporate the updated registry;
5. Empirically re-running the failed-query class against the rebuilt substrate and measuring the HOT rate improvement;
6. Preserving pre-fix misrouting observations and post-fix HOT receipt as Stone Tablet records (per #2327 Stone Tablet Imperative), timestamped with the Call Sign of the repair session.

System Claim

A system for corpus-alias retrieval navigation in a cooperative-platform AI memory infrastructure, comprising:

- A **Cathedral substrate** providing corpus-mode and observational-mode Scribes, each with a primary JSONL corpus and an adjacent-Scribe routing topology;
- A **Scribe-level scoring engine** that selects Scribes by computing keyword overlap between the incoming query and each Scribe's keyword array from the registry;
- A **registry layer** (`registry.yaml`) containing per-Scribe keyword arrays that may be extended with alias phrasings without engine modification;
- A **corpus layer** (per-Scribe JSONL files) containing observation entries written in natural prose with domain-alias vocabulary embedded in observation bodies;
- A **propagation discipline** (this method) that maintains alignment between the alias vocabulary naturally present in observation bodies and the keyword arrays used by

the scoring engine, closing the structural gap as corpus vocabulary evolves;
- An **empirical validation harness** (per-Scribe refire suite) that measures HOT rate before and after alias propagation and preserves the delta as Stone Tablet evidence.

Distinguishing Reduction-to-Practice

The distinguishing technical contribution is the **three-step no-engine-modification discipline**: audit observation bodies → propagate to registry → refire to verify. This contrasts with prior-art approaches that require: (a) retraining the scoring engine on new vocabulary, (b) re-embedding the corpus, or (c) rewriting observation entries to match existing keywords. The registry-keyword-extension method leaves the scoring engine and corpus entries unchanged, achieving alias coverage through the lowest-friction modification path (registry YAML append-only update + rebuild).

Composes With

- #2278 Cathedral Effect — registry-keyword-extension is the navigation discipline within the Cathedral Effect substrate; together they provide model-independent retrieval at 100% HOT empirical receipt
- #2317 Pheromone Substrate — rebuilt keyword matchers are indexed as Pheromone topics
- #2326 Reproducibility Pack — K473 + K547 dual-receipt IS the reproducibility evidence
- #2327 Stone Tablet Imperative — pre-fix and post-fix receipts both preserved as Tablets
- #2332 Wrasse Scribe — registry-keyword additions compose with Wrasse `vocabulary` trigger class for two-layer pre-resolution

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Filing Notes and Counsel Guidance

Innovation Number Confirmation

The following candidate innovation numbers in this filing are **pre-filing assignments** by Bishop (Claude) and require verification against the canonical A&A formal ledger before conversion to utility applications:

Candidate #	Innovation	Source Session
#2327	Stone Tablet Imperative + Vendor-Layer Extension (K541)	B132/K541
#2329	Sweeper Scribe (Triage Primitive)	B132
#2330	Fire Control Directive	B132
#2331	Call Sign Primitive	B132

Candidate #	Innovation	Source Session
#2332	Wrasse Scribe	B132/K540/K544/K545
#2333	Cut In Half / Both-Feet Principle	B132
#2334	Multi-Cylinder Dispatch Patterns	B132
#2335	Registry-Keyword-Extension Method + System	K473→K547/B133

Innovations #2328 (Scavenger Scribe, B131) may require separate filing or inclusion here as the Sweeper (#2329) companion. Counsel to advise on clustering at conversion time.

Innovations #2308, #2311, #2313-#2326 (B127-B131 post-Prov-14 AA formals) are staged in `BISHOP_DROPZONE/12_Innovations_AA/` and are candidates for a companion Prov 15 cluster or the Prov 14 conversion scope. Counsel to advise on optimal clustering.

Stone Tablet Anchors — Call Sign Index

All reduction-to-practice evidence cited in this filing is addressable by Call Sign:

Call Sign	Anchors
`v-vendor-layer-tablet-capture-K-VTC-B132 · d57434b`	#2327 vendor-layer extension
`v-wrasse-scribe-mvp-K540 · 8ca55cd`	#2332 MVP infrastructure
`v-wrasse-wiring-hardening-K544 · 429abbd`	#2332 Phase E conditions 1+2
`v-wrasse-empirical-campaign-K545 · d3802cb`	#2332 41.1% Phase E gate
`v-retrieval-refinement-K547 · 99a4869`	#2335 100% HOT receipt

Filing Gate Status

PUBLICATION GATE: HARD CLOSED

Per Fire Control Directive (#2330): this specification is internal-only pending: 1. Founder review of this document 2. Counsel review (Harrity & Harrity / Lloyd & Mousilli) 3. Founder explicit fire authorization for USPTO submission

Knight scope: build and stage this document. Founder + counsel scope: review and authorize submission.

The mechanical gate flag is: `USPTO_SUBMISSION_AUTHORIZED=false` (this specification).

Filed: K-Prov-15-Spec-Consolidation / B133 / 2026-04-29 by Knight Awaiting Founder + counsel review per Fire Control Directive (#2330)

End of PROV_15_FILING_SPECIFICATION_B133.md